Well-ordering properties: every non-empty subset has a unique least member and the order is irreflexive, asymmetric and transitive

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the least-element conversion, the singleton trick, the three-element transitivity argument and the asymmetry derivation are all verbatim in the span.

Editorial note. The label wo:strictorder names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Definition of a well-ordering. The unit's Well-Orderings section: a well-order is a linear order (connected, transitive, irreflexive-style structure) whose every non-empty subset has a minimal member. Supplies minimality and connectedness used in the first paragraph, and the standing order facts referenced thereafter.

Proposition 0.1. *If $<$ well-orders A, then every non-empty subset of A has a unique $<$-least member, and $<$ is irreflexive, asymmetric and transitive.*

Proof. If X is a non-empty subset of A, it has a $<$-minimal !!element m, i.e., $(\forall z \in X) z \not< m$. Since $<$ is connected, $(\forall z \in X) m \leq z$. So m is the $<$-least !!element of X.

For irreflexivity, fix $a \in A$; the $<$-least !!element of $\{a\}$ is a, so $a \not< a$. For transitivity, if $a < b < c$, then since $\{a, b, c\}$ has a $<$-least !!element, $a < c$. Asymmetry follows from irreflexivity and transitivity $\square$